Harvest Shortfalls, Grain Prices, and Famines in Preindustrial England

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The frequency of bad harvests and price elasticity of demand are measured using new data on English grain yields 1268–1480 and 1750–1850 and a revised price series. The analysis shows that major harvest shortfalls were a significant component of most historical subsistence crises, as back-to-back shortfalls were of the worst famines. Although serious harvest shortfalls long remained an unavoidable fact of economic life, by c.1800 yields had become less variable and prices less harvest sensitive. By the eve of the Industrial Revolution, England had become effectively famine-free.

Was Thomas Robert Malthus correct in his belief that while untold millions of European lives had been blighted by malnutrition, "perhaps in some . . . states an absolute famine may never have been known?"[1] It is a view endorsed by Robert Fogel, who argues that in early modern England and France hunger was commonplace but starvation rare, with scarcities arising more from "failures in the system of food distribution" than from natural calamities *per se*.[2] For Fogel, the real problem was the inelasticity of demand, which magnified modest harvest shortfalls into massive price swings so that staple foodstuffs were rendered unaffordable by those most in need.[3] Karl Gunnar Persson disagrees: he claims that "in early modern Europe, the magnitude of harvest 'defects' was very large indeed."[4] His estimates

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[1] Malthus, *Principle of Population*, Book II, chap. 13.

[2] Fogel, *Escape from Hunger*, pp. 5–7, 53–54. Fogel first proposed this thesis in 1992: "Second Thoughts."

[3] The view that famines might arise as a result of a failure of economic entitlements rather than of food supplies *per se* was first elaborated by Sen, *Poverty and Famines*.

[4] Persson, *Grain Markets*, p. 50 and pp. 47–64.

of the elasticity of demand are higher than Fogel's, implying that distributional dysfunction was more likely to aggravate than cause bad subsistence situations.

Did real supply-side failures pose scant threat to preindustrial England, as Malthus and Fogel both supposed? Or is Persson correct in believing that England was little different from France, Sweden, and other European countries in its periodic exposure to harvest shortfalls, the worst of them big enough to precipitate the heavy excess mortality characteristic of major famines? If the latter, did both the variability of harvests and the price elasticity of demand fall in advance of the Industrial Revolution, so that famine receded as a threat? Or did they fail to change until agriculture and the international trade in foodstuffs were transformed by the technological innovations of the nineteenth century? Hitherto, historians have lacked the combination of good premodern yield and price evidence necessary to address these issues and resolve the contrasting positions of Fogel and Persson. This article redresses that deficiency and analyses the frequency and magnitude of harvest shortfalls and associated price elasticity of demand in England during the three periods 1268–1480, 1750–1850, and 1884–1939.

THE PRICE AND YIELD DATA

For want of reliable early evidence, Fogel and Persson both drew mainly on nineteenth- and early twentieth-century crop yield data —English, French, and Swedish—in their analyses of food supply variability in early modern Europe.[5] Thus Fogel's English grain price elasticity estimates are based on wheat yield data for 1884–1913 and price data for 1540–1840. This article removes the need for such improvisations by combining new high-resolution price and crop yield data sets for late medieval England. The price data set has been constructed by Gregory Clark and incorporates and supersedes earlier ones produced by Henry Phelps Brown and Sheila Hopkins and by David Farmer.[6] The yield data set has been assembled by Bruce Campbell and combines the work of several other historians.[7] Both provide annual data from 1268 until 1480, (when the yield data end).

[5] Fogel, "Second Thoughts"; and Persson, *Grain Markets*, p. 53.

[6] Clark's prices are available at http://www.iisg.nl/hpw/data.php. The correlations between year-to-year differences in the Clark and Farmer ("Prices and Wages," 1988 and 1991) series for 1269–1481 are 0.978 for wheat, 0.920 for barley, and 0.823 for oats. A revised and corrected Phelps Brown and Hopkins crop price series has also been made available by Munro, *Revisions*.

[7] Full details are given in Campbell, "Three Centuries."

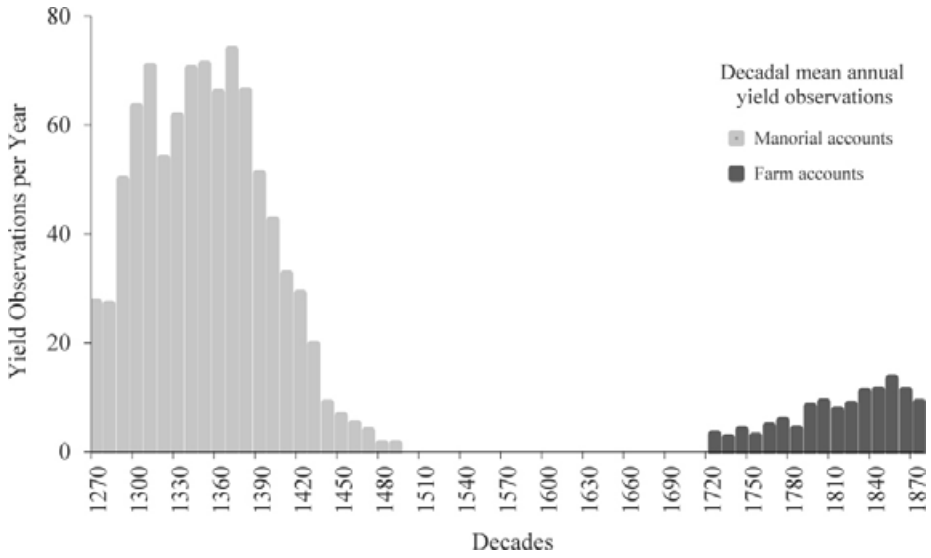


FIGURE 1
MEAN ANNUAL NUMBERS OF GRAIN YIELD OBSERVATIONS FROM MANORIAL ACCOUNTS AND FARM ACCOUNTS

Notes: The 1270s–1490s yield observations are from Campbell, "Three Centuries"; the 1720s–1870s yield observations are from Turner, Beckett, and Afton, *Farm Production.*

Not until the 1750s does an equivalent, though thinner, data set of farm-specific yields become available. These data have been assembled from farm accounts by Michael Turner, John Beckett, and Bethanie Afton and extend to the mid-nineteenth century (Figure 1).[8] To facilitate comparison between the two yield datasets and control for changing patterns of geographical coverage, national time series have been reconstructed using a common methodology and system of regional weights.[9] The period from 1480 to 1750 could not be covered because the available sources on yields (probate inventories or tithe records) do not provide a direct and price-independent measure of harvest quality.[10]

[8] Turner, Beckett, and Afton, *Farm Production*.

[9] The data sets of farm- and year-specific yields are subdivided into seven regional groupings to create a separate series for each of these regions. The regional series were derived using regression analysis with dummy variables for each farm and year, as suggested by Clark, "Price History." Because the dispersion of grain yields across farms is very high, a log-linear specification has been used. Deduction of tithes at source has been assumed to have been a consistent 10 percent of the gross harvested crop. These are then combined into a single national master series by weighting each region according to its share of the national sown area in 1801 (for details, see Apostolides et al., "English Agricultural Output").

[10] Overton, "Re-estimating Crop Yields"; and Apostolides et al., "English Agricultural Output." Short-term fluctuations in tithe receipts could occur for a variety of institutional reasons: tithes might be leased out, recipients might forgo or reduce payments in bad year, or taxpayers might refuse to pay: Dodds, "Estimating Arable Output." Examples of rich, high-quality tithe series are

With its continuous span of more than 200 years, the medieval crop yields data set is the single largest body of English yield observations prior to the inclusion of yield information in the official agricultural statistics from 1884 (Figure 1). It contains almost 30,000 individual yield observations for over 200 seigniorial demesnes concentrated within the core regions of grain production in southern and eastern England. Data are fullest for the period c.1280 to c.1430, when making and keeping annual manorial accounts was most widespread and before leasing had become the almost universal practice of landlords. The raw data provide precisely dated yields per seed and yields per unit area of specific grains on specific demesne farms. These were invariably large-scale enterprises, which differed from smallholdings in terms of their cultivation techniques, capital investment, crop choice, and labor process. Nevertheless, although absolute levels of yield may have been different between the seigniorial and non-seigniorial sectors, high correlations between demesne yields and aggregate tithe receipts from all grain producers indicate that crop-specific year-to-year fluctuations in yields were broadly similar under both regimes.[11] It should be emphasized that these data relate exclusively to the rates of return per units of seed and land sown. Output variation arising from changes in the area under crop is not observable.[12]

THE FREQUENCY AND SCALE OF BAD GRAIN HARVESTS

Beginning with W. G. Hoskins's pioneering attempt to infer harvest quality from English grain prices, historical analysis of "bad years" has typically focused upon wheat, the most commercialized and consequently best recorded of the major grains.[13] This, however, fails to do full justice to the agronomic reality, for oats and barley were also significant crops in medieval and early modern England. Both were staple ingredients of the coarser and cheaper grades of bread and were widely consumed as pottage by the poorer classes. Hence they are fully relevant to issues of subsistence. Moreover, both were better suited to brewing than wheat, and oats in particular was a staple fodder crop. Together, by value they consistently accounted for between a third and a half of all grain produced between 1300 and 1850.[14]

given in Head-Koenig and Veyrassat-Herren, "Les revenues décimaux"; Dodds, *Peasants and Production*; and López-Salazar Pérez and Galán, "La producción cerealista."

[11] Campbell, "Four Famines."

[12] Campbell, *English Seigniorial Agriculture.*

[13] Hoskins, "Harvest Fluctuations," pp. 29–30.

[14] From 1300 to 1850 barley and oats consistently accounted for 32–48 percent by value of

Whereas wheat was invariably sown in the autumn, most barley and virtually all oats were sown in the spring. These spring grains were therefore largely unaffected by a bad autumn or winter and might even be sown as a replacement for wheat or rye if the latter seemed to have been severely damaged by harsh winter conditions. Since these grains differed in their nutrient demands upon the soil, they also formed complementary components of the more elaborate rotations as exemplified by the classic three-course rotation of winter crops, spring crops, and then a full year's recuperative fallow. Of these three staple grains, wheat may have been the most expensive and widely traded but only oats was capable of being grown in virtually all farming regions since it was tolerant of the widest range of growing conditions. In fact, in many upland areas it was almost the only grain grown.[15] To overlook oats is therefore to ignore the nation's most universal cereal. Oats was also far more affordable than wheat and its harvest was consequently more material to the nutrition of the poor.

Sowing wheat in the autumn and almost all oats in the spring offered a good deal of insurance to cultivators and consumers, as captured by the relatively low, albeit positive, correlation between wheat and oats yields (Table 1). The correlation between first differences in net yields was +0.06. As is to be expected, year-to-year changes in oats and barley yields were more highly correlated (+0.30). The yield variability of a composite cereal (W-B-O) based on weights of 0.5 for wheat and 0.25 each for barley and oats (weighting the winter- and spring-sown grains equally and, within the spring sector, barley and oats equally) was considerably less than that of wheat alone.[16] Evidently, growing and harvesting conditions had to be exceptionally good or bad for the yields of winter and spring grains to have been similarly affected. This mattered little when both did well, but a lot when both fared badly, since in these cases cultivating both types of grain provided no insurance. Tellingly, the correlation coefficient between oats and wheat yields reaches +0.391 over the watershed period 1314–1351 when positive and negative environmental forcing of harvests was exceptionally strong, compared to only +0.169 over the entire period 1269–1480.[17]

national grain output net of seed and fodder and 42–58 percent by volume: Broadberry et al., "British Economic Growth."

[15] Collins, "Dietary Change;" and Overton, "Agriculture," pp. 37–41.

[16] See footnote 14. These weights are necessarily somewhat arbitrary, but reflect the primacy of wheat noted in the text. The standard deviation of the unfiltered wheat yield series for the 1270–1475 period is 22.6; that of W-B-O is 17.8, or nearly 18 percent less.

[17] Campbell, "Physical Shocks."

TABLE 1
ENGLAND, 1268-1480 AND 1750-1850, BETWEEN-CROP YIELD AND PRICE CORRELATIONS

	Between-Crop Correlation Coefficients:					
A: 1268–1480	Net Yields (1268–1480)			Prices (1269–1481)		
	Wheat	Barley	Oats	Wheat	Barley	Oats
Wheat	1.000			1.000		
Barley	0.357	1.000		0.793	1.000	
Oats	0.169	0.416	1.000	0.744	0.850	1.000
B: 1268–1349	Net Yields [1268–1349]			Prices [1269–1350]		
	Wheat	Barley	Oats	Wheat	Barley	Oats
Wheat	1.000			1.000		
Barley	0.403	1.000		0.956	1.000	
Oats	0.284	0.384	1.000	0.839	0.862	1.000
C: 1350–1480	Net Yields [1350–1480]			Prices [1351–1481]		
	Wheat	Barley	Oats	Wheat	Barley	Oats
Wheat	1.000			1.000		
Barley	0.314	1.000		0.676	1.000	
Oats	0.154	0.436	1.000	0.661	0.849	1.000
D: 1750–1850	Gross Yields [1750–1850]			Prices [1751–1851]		
	Wheat	Barley	Oats	Wheat	Barley	Oats
Wheat	1.000			1.000		
Barley	0.409	1.000		0.939	1.000	
Oats	0.393	0.555	1.000	0.914	0.961	1.000

Note: The yield data are from Campbell, "Three Centuries"; and Turner, Beckett, and Afton, *Farm Production*; the price data are from http://www.iisg.nl/hpw/data.php.

As Table 1 reveals, movements in the prices of the three crops were normally much more highly correlated, suggesting the integration of markets across, and interchangeability of demand between, cereals.[18] By the period 1750–1850 all correlations are higher but the wide differential between intercrop correlations of yields and prices nonetheless remained.

[18] Between 1271 and 1476 the correlations between price and net yield were –0.61 for wheat (the price maker), –0.24 for barley, and –0.37 for oats. The correlations between first differences were –0.62, –0.31, and –0.20, respectively. On market integration in this period, see Clark, "Markets and Economic Growth."

For all grains, yield variability seems to have shifted after 1350, although not in a simple or consistent way (Figures 2A and 2B). Wheat yields were most unstable in the half-century 1350–1399, when a succession of below-average harvests from 1350 to 1375 was followed by a run of the most bountiful harvests on record. In contrast, barley and oats yields both fluctuated most after 1450, when the data are thinnest (Figure 1) and least representative and the accuracy with which harvests were recorded may be declining.[19] These differences also reflect the subtle but profound shifts in weather conditions highlighted by recent environmental research.[20] Yet climate differences alone can hardly account for the fact that, as reconstructed from farm-level data, medieval yields of wheat and barley were three times, and medieval oats yields more than four times, as variable as modern (post-1884) English/British yields documented by the official agricultural statistics.[21]

Across the entire period 1268–1480, using gross yields per acre, the detrended data indicate that the probabilities of a harvest shortfall of one-tenth or more in a given year were 0.23 for wheat, 0.22 for barley, and 0.19 for oats.[22] Using yields net of seed, the respective probabilities were 0.27, 0.29, and 0.26. The probabilities of a far greater shortfall of one-fifth or more were significantly lower: for wheat, barley, and oats, they were 0.10, 0.08, and 0.08 (gross), and 0.17, 0.15, and 0.12 (net). The insurance offered by sowing a range of crops is indicated by shortfall probabilities for W-B-O: for yield shortfalls of one-tenth and one-fifth, these were 0.16 and 0.05 (gross) and 0.22 and 0.11 (net). In other words, on average, one grain harvest in nine delivered a net yield at least 20 percent below trend. For wheat, however, shortfalls of this magnitude occurred at least once every six years. Prudent farmers therefore spread their risks and grew a mix of winter- and spring-sown grains.

Historically, the worst famines as well as the episodes of most acute dearth have been the product of back-to-back harvest failures.[23] Considered as random events, such repeat failures are bound to have

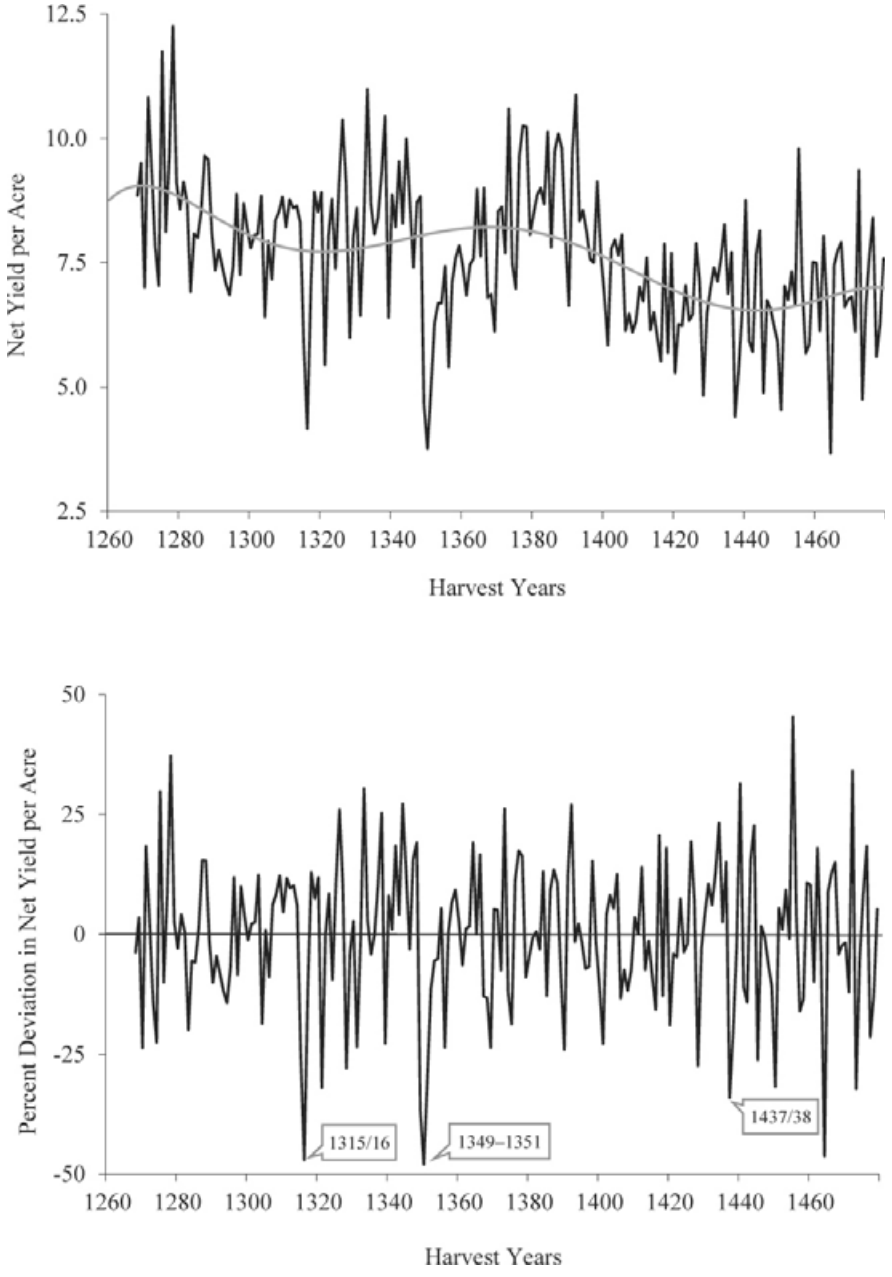
[19] Campbell, "Grain Yields on English Demesnes."

[20] Dawson et al., "North Atlantic Climate Changes"; Trouet et al., "Persistent Positive North Atlantic Oscillation"; and Campbell, "Panzootics" and "Grain Yields on English Demesnes."

[21] See Table 3.

[22] Detrended using a Hodrick-Prescott filter.

[23] "Great" famines which have resulted from such consecutive harvest failures include the "Great" European Famine of 1315–1317, the "Great" Irish Famine of 1845–1850, the "Great" North China Famine of 1876–1878, and the "Great" Leap Forward famine of 1959–1961: Ó Gráda, "Making Famine History."



FIGURES 2A and 2B
ENGLAND, 1268–1480, DEVIATIONS FROM TREND IN NET WHEAT-BARLEY-OATS (W-B-O) YIELDS

Note: The yield data are from Campbell, "Three Centuries." W-B-O is a weighted average of wheat, barley, and oats yields, using the ratio 2:1:1.

been rare, but there are three reasons why harvest outcomes might be serially correlated. First, there could be some autocorrelation in environmental conditions, as in the case of the major climate anomaly responsible for the Great European Famine of 1315–1321 and concurrent European cattle panzootic of 1316–1321, and/or plant diseases such as the repeated mildew attacks recorded in Kent between 1365 and 1375 or, most notoriously, the blight-induced Irish Potato Famine of 1845–1851.[24] Second, a poor harvest, indicated by a low yield ratio, might mean thinner sowings and deficient yields or smaller sown areas and diminished output the next year (note, on the demesne farms represented in the medieval crop yields database, it was the seeded area rather than seeding rate that was most likely to be sacrificed). Third, any reductions in fertilizer and draught power following a bad year, whether because animals were dead or malnourished, or because farmers themselves were suffering, could reduce yield ratios. It turns out, however, that autocorrelations in yields, though generally positive, were small and never statistically significant either in the medieval period or in 1750–1850.[25]

The top two panels of Table 2 are derived from fitting the medieval yield data to a Hodrick-Prescott filter (which extracts the cyclical component from a time series) and identifying those years with gross and net yields at least 10, 20, and 30 percent below trend. Focusing on net yields, between 1268 and 1480 back-to-back shortfalls of over 10 percent for individual cereals were not unusual. They were also somewhat more common than might be expected on the basis of squaring once-off probabilities. For example, the statistical probability of a back-to-back shortfall in net wheat yield of 10 percent or more was one in twelve (0.085), but such shortfalls occurred 14 percent more often than "expected." Note that a serious shortfall was less likely for oats, the buffer crop of the poor, than the other two cereals. Unsurprisingly, the chance of all three major cereals simultaneously doing badly was lowest of all: the probability of a 10 percent back-to-back shortfall in the net W-B-O yield was one in fifteen (0.066). A failure on such a scale across all crops would almost certainly have led to genuine hardship and malnutrition for many, though hardly classic famine conditions. Back-to-back shortfalls sufficient to result in major famines were much rarer, especially those affecting all three major grains. For wheat and barley,

[24] Campbell, "Agriculture in Kent," p. 47.

[25] Between the Middle Ages and 1750–1850 the autocorrelation of wheat yields switched from positive to negative; those of barley, oats, and W-B-O remained positive while decreasing in size.

TABLE 2
LIKELIHOOD OF "BACK-TO-BACK" HARVEST FAILURES, ENGLAND, 1268–1480 AND 1750–1850

Back-to-Back Deficits, 1268–1480	Wheat	Barley	Oats	W-B-O
Prob < 90% of trend value	0.085	0.089	0.075	0.066
Average/Expected	1.140	1.088	1.127	1.350
Prob < 80% of trend value	0.033	0.042	0.014	0.019
Average/Expected	1.089	1.872	0.945	1.479
Prob < 70% of trend value	0.019	0.009	0.005	0.009
Average/Expected	2.360	1.664	0.832	6.656
Three-in-a-Row Deficits, 1268–1480	**Wheat**	**Barley**	**Oats**	**W-B-O**
Prob < 90% of trend value	0.005	0.014	0.000	0.005
Average/Expected	0.47	1.60	1.64	1.31
Prob < 80% of trend value	0.005	0.014	0.000	0.005
Average/Expected	0.90	4.15	0.00	3.28
Prob < 70% of trend value	0.005	0.000	0.000	0.005
Average/Expected	6.62	0.00	0.00	88.61
Back-to-Back Deficits, 1750–1850	**Wheat**	**Barley**	**Oats**	**W-B-O**
Prob < 90% of trend value	0.040	0.079	0.069	0.040
Average/Expected	0.84	1.67	0.84	1.12
Prob < 80% of trend value	0.00	0.010	0.040	0.00
Average/Expected	0.00	1.01	5.00	0.00
Prob < 70% of trend value	0.00	0.00	0.020	0.00
Average/Expected	—	—	0.62	—

Notes: The W-B-O column is a weighted average of the net cereal yields of wheat, barley, and oats, using the ratio 2:1:1. "Average/Expected" is the ratio of the actual to predicted frequency. The predicted frequency is set at the once-off probability of such a shortfall squared or cubed. The yield data are from Campbell, "Three Centuries," and Turner, Beckett, and Afton, *Farm Production*, (the annual series have been reconstructed from original data supplied by the authors).

consecutive net reductions of a fifth or more occurred only about once every three or four decades, although in this period they happened somewhat more often than might be expected if harvests were purely random (as indicated by the values of the average/expected entries in Table 2). The most dangerous shortfalls of all were double back-to-back (i.e., three-in-a-row) failures of this magnitude, but they were "out-of-the-blue" events likely to occur once in every two or three centuries.[26] Evidence on their randomness is mixed; they were more frequent in the case of wheat (gross yield), barley (net yield), and W-B-O (net yield), but not otherwise (Table 2).

[26] Only in the case of net barley yields are any of the entries in the bottom half of Table 2 greater than 0.005.

Across the entire period, 1268–1480, three episodes registered back-to-back reductions in the net W-B-O yield in excess of 25 percent (Figures 2A, 2B, 3A, and 3B): 1315/16, 1349–1351 (the only three-in-a-row failure of this magnitude), and 1437/38. In 1315 and 1316 the yield reductions exceeded 25 and 47 percent, respectively, and they exceeded 30 percent in 1349, 1350, and 1351, and also in 1437. The greatest yield reductions of all occurred in 1316 (47 percent), 1350 (47 percent), 1464 (46 percent, but based upon little data), and 1349–1351 (over 30 percent in all three years) and provide a yardstick against which all other serious shortfalls can be measured.[27] In climatically temperate England, this was as bad as it could get. The years 1315 and 1316 thus stand out as having witnessed the sharpest and deepest yield reduction during the period under review, with winter-sown wheat and rye exceptionally hard hit, and 1349–1351 the most prolonged, when, although wheat again fared worst, deficient crops of barley and oats greatly exacerbated the crisis.

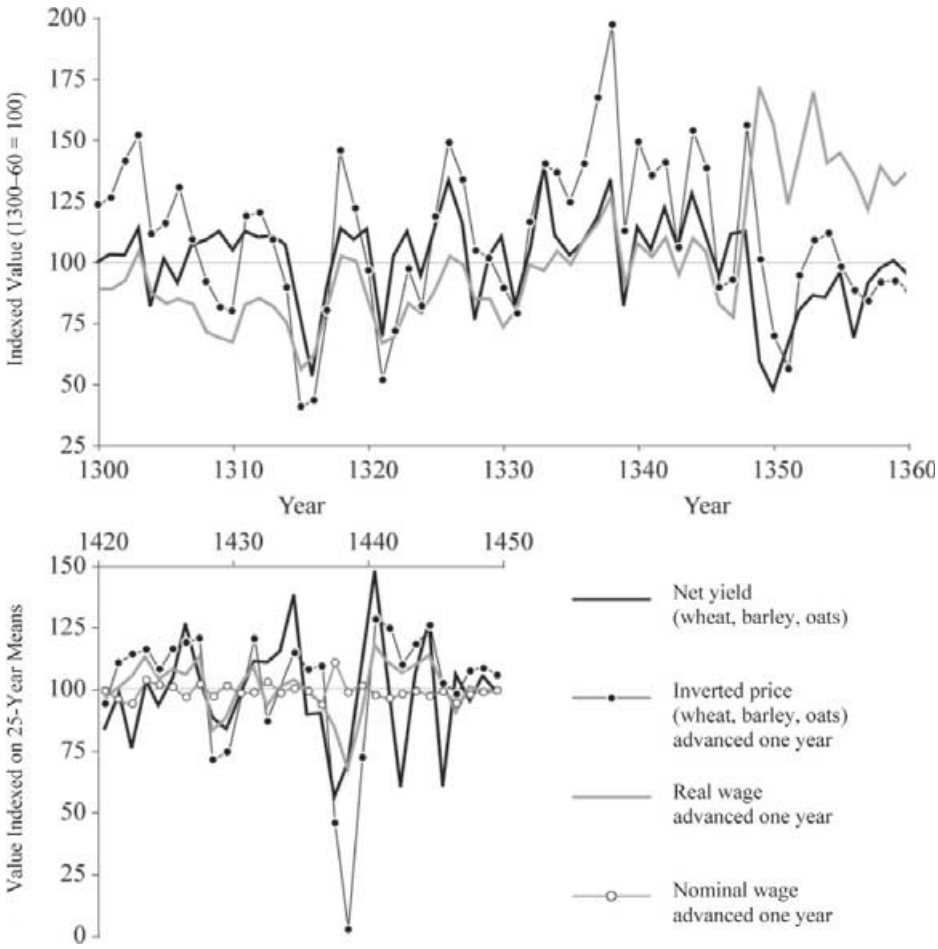
Other things being equal, of course, the disastrous three- and four-year failure of 1349–1351/52 should have presented producers and consumers with a greater challenge of survival than the marginally more extreme two-year failure of 1315/16. That, paradoxically, this was not the case is brought out by the movement of prices and wages. Figures 3A and 3B plot the combined net yield per seed of W-B-O, a combined price for these same three grains (inverted and advanced one year to bring out the coincidental movement between yields and prices), and Clark's farmworker's real wage rate (also advanced one year). In a period notable for the variability of its yields, the pronounced back-to-back supply-side shocks of 1315/16 and 1349–1351 stand out clearly. Not only was the latter the more prolonged event, recovery from it was also slower, since it was not until the late 1370s that yields were restored to the levels recorded on the eve of the crisis.[28] Nevertheless, it was the harvest failure of 1315/16 which triggered by far the greater and more immediate price response, initiating an inflation of grain prices that was as unparalleled as it was unprecedented.[29] In contrast, even making allowance for the substantial reduction which had taken place in the amount of money in circulation, the price response to the 1349–1351 shortfall was slower and disproportionately slighter and it was not until the second successive serious failure that prices belatedly began to rise.[30]

[27] In 1464 the wheat crop was just below trend, but the oats and barley crops were extremely poor.

[28] Campbell, "Grain Yields on English Demesnes."

[29] Campbell, "Nature as Historical Protagonist," pp. 285–86.

[30] Allen, "Silver Production," p. 128, estimates that the value of the English currency declined from £1.8–2.3m. (+ foreign gold) in 1319 to £0.6–0.95m. (+ foreign gold) in 1351. After allowance



FIGURES 3A and 3B
ENGLISH HARVESTS, GRAIN PRICES, AND FARMWORKERS' REAL WAGES DURING THE CRISES OF 1315–1317, 1349–1351, AND 1437/38

Note: The yield data are from Campbell, "Three Centuries"; the price data are from http://www.iisg.nl/hpw/data.php; and the wage data are from Clark, "Long March of History," pp. 132–33. The yields and prices are combined using the formula: (wheat x 0.5) + (barley x 0.25) + (oats x 0.25). The prices and wages are advanced one year so that they synchronize with the yields. Prices are inverted because the normal response to a reduction in yield was a rise in price.

During the earlier of these two crises, the scarce grain, reduced agricultural employment, and sky-high food prices had squeezed incomes of wage earners hard, and the real wage rate of English

for the mortality of the Black Death, money supply per capita was approximately 40 percent lower in 1351 than it had been in 1319.

farmworkers sank to its lowest recorded point ever.[31] Excess mortality probably thinned the ranks of laborers, as is implied by the modest rise in nominal wages in the wake of the crisis.[32] Contemporary accounts leave no doubt that England and much of northern Europe were in the grip of dire famine.[33] In addition, all the classic economic symptoms of famine are observed during these grim years: rural credit arrangements collapsed triggering a wave of petty land sales, and many also traded parcels of land for food; crimes against property soared, as individuals were driven to seek food by any means; rural-to-urban migration increased in a desperate quest for food and alms; marriages were postponed for want of the means to establish new households; and there are clear signs that death rates rose.[34]

That the consecutive harvest failures of 1315 and 1316 resulted in a genuine famine with great economic hardship and significant excess mortality is clear. Why, then, did the three-in-a-row failure of 1349–1351, so much greater in its cumulative incidence, not produce a subsistence crisis of similar or superior magnitude? The explanation lies in its coincidence with an even greater demand-side shock: the Black Death. In fact, the massive mortality of workers and managers —reckoned to have been in excess of 40 percent—is likely to have compounded the harvest failure and may well explain why on this occasion the cheaper spring grains of barley and oats together with peas and other legumes fared particularly badly, since priority was given to harvesting and threshing the more highly valued wheat.[35] Had this been an "ordinary" harvest failure—as in 1315/16 and again in 1437/38 (Figures 3A and 3B)—farmworkers' real wages should have collapsed. Instead, in 1349 real wage rates rose dramatically and, paradoxically, remained high in 1350 notwithstanding a second consecutive bad

[31] The lowest point on Clark's 1209–1869 index of farmworkers' real wage rates is 1316 ("Long March of History," pp. 130–34), whereas 1597—another year of extreme scarcity—is the lowest point on the Phelps Brown and Hopkins 1264–1954 index of building workers' real wage rates (*Wages and Prices*, pp. 28–31).

[32] Clark, "Long March of History," p. 132, reports an 18 percent gain in nominal wages between 1314 and 1322. Razi, *Life, Marriage, and Death*, pp. 24–45, and Poos, "Rural Population of Essex," document significant declines in numbers of adult males across this same period.

[33] Jordan, *Great Famine.*

[34] Kershaw, "Great Famine"; Jordan, *Great Famine*; and Campbell, "Nature as Historical Protagonist," pp. 293–96.

[35] Chronicler Henry Knighton noted that "many crops rotted in the fields for want of harvesting . . .there was such an abundance of grain that almost no one cared for it" (*Knighton's Chronicle*, p. 101), although this description probably relates more to the situation on peasant holdings than on demesnes. For harvesting problems on the Kent demesnes of Canterbury Cathedral Priory, see Campbell, "Agriculture in Kent," pp. 46–47.

harvest.[36] For the real wage rate of the most economically marginal group to have registered such striking gains during one of the worst back-to-back harvest failures on record highlights the unique nature of this, the greatest (in relative terms) of all major historical mortality crises. Famine on this occasion had been forestalled by plague, and the windfall economic gains bestowed upon survivors of the pandemic and its repeated visitations meant that it would be a long time before England's population would again be so vulnerable to harvest failure.

Although the Black Death is the most dramatic example of the capacity of epidemic disease to transform a potentially dire subsistence situation into a crisis of an entirely different nature and order, historically it is not unique.[37] On this occasion, however, the abject poverty of such a large proportion of the population clearly worsened the situation. Incomplete economic and nutritional recovery from the earlier back-to-back harvest failure of 1315–1317 and its ensuing cattle panzootics and Scottish raids further exacerbated the situation, especially in the worst affected regions.[38] It was fortunate, therefore, that the next serious back-to-back harvest failure of 1437/38 (Figure 3B) occurred at a time when population levels were low, poverty much reduced, and the purchasing power of wage earners substantially improved.[39] Once again, it was prevailing socioeconomic conditions which largely determined the impact of, and responses to, a major exogenously determined food-availability decline. Rural prosperity had certainly not granted immunity to the consequences of harvest failure

[36] Clark's farmworker's wage series is used since it captures best the improved circumstances of workers lucky enough to have survived. Labor's gain relative to land is captured in Henry Knighton's lament that "ox hides fell to a wretched price, namely 12d., and yet a pair of gloves would cost 10d., 12d., or 14d., and a pair of breeches 3s. or 4s." (*Knighton's Chronicle*, pp. 103, 105) Clark demonstrates that Edward III's efforts at regulating wages through the Ordinance of Labourers (1349) and the Statute of Labourers (1351) had only limited effect ("Long March of History," pp. 115–17). Penn and Dyer ("Wages and Earnings," p. 359) state that those laws were "broken in each year by hundreds of thousands of workers." A slump in construction activity nevertheless meant that building laborers failed to make comparable gains in real wages: Munro, "Before and After the Black Death."

[37] For other English examples, see Campbell, "Four Famines." An example, familiar to readers of Alessandro Manzoni's 1827—revised 1842—novel, *The Bethrothed (I Promessi Sposi)*, is the plague that followed famine in northern Italy in 1629/30.

[38] Campbell, "Nature as Historical Protagonist" and "Physical Shocks." Probably 30–40 percent of households lived at the most basic of subsistence levels: research into the proportion of households able to afford no more than a bare-bones basket of consumables forms one strand of the Leverhulme-funded research project (Ref: F/00215AR), "Reconstructing the National Incomes of Britain and Holland, *c.*1270/ 1500 to 1850," directed by Stephen N. Broadberry, Bruce M. S. Campbell, Mark Overton, and Jan Luiten van Zanden, with research assistance by Alex Klein and Bas van Leeuwen.

[39] By this date, households subsisting on a bare-bones basket of consumables comprised less than a fifth of the total (see footnote 38).

but it greatly mitigated the hardship that resulted. It follows that the risk of serious famine was greatest at times of greatest economic austerity, as in 1315–1317 and 1349–1351.[40]

DECLINING YIELD VARIABILITY OVER TIME?

Did the high variability of medieval yields persist into modern times? The bottom panel of Table 2 above, which reports the likelihood of back-to-back deficits in 1750–1850, shows, that by the eighteenth century, things had changed. Table 3 addresses the question more directly, by comparing the coefficients of variation calculated on detrended annual series of *gross* wheat and barley yields for the three periods 1268–1467, 1750–1850, and 1884–1939, and the coefficients of variation of oats yields for 1268–1480 and 1884–1939.[41] Significantly, the table highlights a key unsung achievement of English farmers and their seed suppliers between the Middle Ages and the late eighteenth century. The reduction in the year-to-year variability of yields is substantial compared to 1268–1467, the coefficients of variation for wheat and barley are respectively a quarter and an eighth lower in 1750–1850. Unfortunately, the lack of intervening data means little can be said about the timing of the reduction in yield variability. Given that the medieval data employed here are "thicker" than those for 1750–1850 (Figure 1), the comparison, if anything, understates the achieved reduction in variability.

The claim that yield variability fell in the early modern era is not entirely novel. Over three decades, ago Hugues Neveux and Anne-Marie Tits-Dieuaide concluded, on the basis of an analysis of Flemish and southern English yields, that yield variability on the eve of World War II was only a third what it had been in the later Middle Ages.[42] Certainly, the range of solutions proposed in print to crop damage caused by rain, fungi, weeds, and pests grew from the sixteenth century onwards.[43] Fogel and Persson's claim that

[40] Compare also the situation in the 1590s: Campbell, "Four Famines," pp. 36–47.

[41] Turner, Beckett, and Afton (*Farm Production*, p. 163) base their crop yields for wheat between 1700 and 1850 on 829 observations, but those for oats on only 444 observations (or an average of less than three per year). Given the small number of underlying observations and the implausibly huge year-to-year variations in gross yields that they imply, oats have been excluded from consideration in 1750–1850.

[42] Neveux and Tits-Dieuaide, "Étude structurelle des fluctuations courtes." The claim is based on an analysis of data in Titow, *Winchester Yields*, and Tits-Dieuaide, *La formation des prix céréaliers.* More recently Jean-Marc Moriceau (*Terres Mouvantes*, pp. 243–44 and *Fermiers*, p. 428) has linked a likely, but undocumented, decline in crop yield variability in France to "*des mutations génétiques*," stemming from gradually improving seed selection.

[43] Mokyr, *Lever of Riches*, pp. 32, 296; Campbell, *English Seigneurial Agriculture*, pp. 414–15;

TABLE 3
ENGLAND/GREAT BRITAIN, ANNUAL VARIATIONS IN GROSS GRAIN YIELDS

Crop	Period	Country	(1) Mean	(2) σ	(2)/(1) CV
Wheat	1268–1467	England	9.79	1.60	0.165
Wheat	1750–1850	England	22.54	2.81	0.125
Wheat	1884–1939	Great Britain	31.26	2.23	0.071
Barley	1268–1467	England	13.18	2.00	0.151
Barley	1750–1850	England	33.34	4.35	0.131
Barley	1884–1939	Great Britain	33.00	1.79	0.054
Oats	1268–1467	England	10.14	1.47	0.145
Oats	1750–1850	England	NA	NA	NA
Oats	1884–1939	Great Britain	40.14	1.81	0.045
W-B-O	1268–1467	England	10.72	1.19	0.111
W-B	1750–1850	England	27.94	2.72	0.097
W-B-O	1884–1939	Great Britain	33.92	1.68	0.050

Notes: The standard deviations were estimated after the chronologies had been detrended using the Hodrick-Prescott filter. CV is the coefficient of variation. W-B-O is a weighted average of wheat, barley, and oats yields, using the ratio 2:1:1. W-B is a weighted average of wheat and barley yields, using the ratio 1:1. The yield data are from Campbell, "Three Centuries"; Turner, Beckett, and Afton, *Farm Production* (the annual series have been reconstructed from original data supplied by the authors); and Mitchell, *British Historical Statistics*, pp. 90–91.

postmedieval technological change greatly improved grain yields but had little impact on their variability may therefore be incorrect.[44] On the evidence of Table 3, a modest lowering in the coefficient of variation of wheat yields from 0.165 between 1268–1467 to 0.125 between 1750–1850 would have meant a reduction from roughly 12 percent to 5 percent in the chance of harvests falling short in any year by a fifth or more.[45] The socioeconomic implications of such an improvement would have been significant. It likely also made a significant contribution to the postmedieval marked dampening down of price fluctuations and must be one of several reasons why famines in England became less common. Moreover, comparison with a corresponding set of coefficients calculated using national output statistics for the period 1884–1939 (Table 3) implies that yield variability declined further during the nineteenth century, although the differing construction of the two data sets may exaggerate the scale of that decline.[46]

Allen, *Enclosure and the Yeoman*, pp. 206–07; Turner, Beckett, and Afton, *Farm Production*, pp. 78–81; Buttress and Dennis, "Seed Treatment"; Smith and Secoy, "Salt as a Pesticide"; and Thick, "Garden Seeds in England."

[44] Fogel, "Second Thoughts," p. 252, fn. 8. Persson (*Grain Markets*, p. 113) also believes that between the sixteenth and nineteenth centuries "while yields increased there is to my knowledge no evidence that yield variance changed much."

[45] In only one instance between 1700 and 1820—in 1719/20—did the wheat crop suffer a back-to-back shortfall of over 20 percent below trend, and there was no case of a back-to-back shortfall as great as 30 percent.

[46] At issue is the level at which yields are calculated; for the periods 1268–1467 and 1750–

HARVESTS AND PRICES

Investigation of the relationship between proportionate harvest shortfalls and resultant increases in the price of grain began with Charles Davenant's celebrated table purporting to describe the market for wheat in seventeenth-century England.[47] The schedule implies a price elasticity of demand of –0.403.[48] Since Davenant lacked aggregate agricultural data to build upon, his table must be considered no more than at best an informed conjecture. Fogel, working from non-coterminous yield and price data has proposed a substantially lower elasticity figure of –0.183 for early modern England, from which he infers that very high prices in Europe in the early modern era did not result from huge harvest shortfalls. In the specific case of England, he concludes that "there does not appear to have been a single year after *c*.1500 in which aggregate supply of food was too low to avoid a subsistence crisis."[49] He further buttressed his case for a low demand elasticity with demographic data from E. A. Wrigley and R. S. Schofield, which suggested that mortality in preindustrial England was high but not subject to large year-to-year fluctuations.[50] Since low elasticities meant that high prices were generated more by poor access to, than inadequate supplies of, grain, scope was left for public action to redistribute available food stocks. This led Fogel to argue (in the spirit of Amartya Sen's *Poverty and Famines*) that any major European subsistence crises and famines in the early modern era were less due to food-availability declines than to distributional deficiencies, including public inaction or misgovernance.

Rafael Barquin Gil has replicated Fogel's method and data and revised the result upwards to –0.36, which is fairly close to the value implied by Davenant.[51] Also using the same method but working from French and Swedish data, Persson has derived an even higher

1850 this is the individual farm, but in 1884–1939 it is undoubtedly at a higher but unspecified level of spatial aggregation. For the latter reason, macro-level agricultural statistics can be a poor guide to variations in output at the micro-level, where the difference between sufficiency and scarcity really counted.

[47] It is often incorrectly ascribed to Gregory King.

[48] Stigler, *Essays*, p. 215, fn. 32. Philip Wicksteed (see Stigler, *Essays*, p. 214, fn. 30) found that the equation $60p = 1{,}500 - 374q + 34q^2 - q^3$ where normal q is 10 and normal p is 1, provided an exact fit for Davenant's schedule. Wrigley ("Reflections on Corn Yields," pp. 92–95), reports two other formulations, by Jevons and Bouniatian. As Fogel ("Second Thoughts," p. 252) notes, these estimates are "true" elasticities only in the absence of storage and trade (see footnote 58).

[49] Fogel, "Second Thoughts," p. 255; but compare Appleby, *Famine*; Hoyle, "Famine"; and Campbell, "Four Famines."

[50] Wrigley and Schofield, *Population History*.

[51] Barquin, "Elasticity of Demand," p. 251.

elasticity estimate of –0.6.[52] Price elasticities of this magnitude imply that serious harvest shortfalls were a reality in early modern Europe, and so resuscitate the historical significance of subsistence crises and rationalize the need for proactive public policy in a context where markets still functioned slowly and natural hazards sporadically had a major impact upon the output of staple foodstuffs.

In late medieval England, price elasticities for wheat calculated using the data set of yields described above were even higher than those obtained by Persson and Barquin using postmedieval data. Thus, applying the elasticity measure (Method 1) $\varepsilon = -\sigma_q/\sigma_p$ employed by Fogel [where ε is the price elasticity of demand, and the σs are the standard deviations of output (q) and price (p)] to the coefficients of variation of net wheat yield and price for the years 1268–1480 produces an elasticity estimate of –0.73 for wheat.[53] Using gross yields generates the lower value of –0.55 (Table 4).[54] These estimates are substantially above those based upon nineteenth-century Swedish and French data offered by Persson and higher, too, than Barquin's revised estimate for early modern/nineteenth-century Britain.[55] They also exceed an estimated elasticity of –0.35 obtained by applying the same method to Turner, Beckett, and Afton's gross wheat yield data for 1750–1850 (Table 4). In short, they imply that in preindustrial England at least price elasticities were far from constant and tended to decline over time.

A simple alternative method of calculating price elasticities when coterminous annual price and yield data exist may be obtained by estimating equation (Method 2):

$$\ln(q_t) = a + b\ln(p_t)$$

where p and q are price and yield, a is a constant term, and b is the elasticity to be estimated.[56] Estimating this equation in first differences

[52] Persson, *Grain Markets*, p. 54.

[53] Both the yield and price series have been detrended using a Hodrick-Prescott filter. Deviations from trend are used to calculate the standard deviation.

[54] Equivalent elasticities for barley and oats are –0.79 and –0.87 (net) and –0.57 and –0.61 (gross).

[55] Barquin, "Elasticity of Demand," pp. 263–64. Barquin also estimated elasticities of –0.8 for wheat from French and Swiss tithe data spanning the sixteenth to eighteenth centuries.

[56] Although an increase in storage and in trade would cushion the impact of any given harvest on price, thus yielding lower elasticities, how much prices responded to domestic quantity shocks is still an important issue, since it is a measure of the vulnerability of net consumers of cereals to local harvest shortfalls. Clearly, more inventory adjustment and greater market integration would reduce measured "elasticities." We are grateful to one of the referees for insisting on this point.

TABLE 4
ENGLAND, 1268–1480 AND 1750–1850, ESTIMATED WHEAT PRICE ELASTICITIES OF DEMAND

Period	Wheat Price Elasticity (Method 1)	Wheat Price Elasticity (Method 2)
1268–1480 (net yield)	–0.73	
1268–1480 (gross yield)	–0.55	–0.57
1750–1850 (gross yield)	–0.35	–0.23

Notes: The yield data are from Campbell, "Three Centuries"; and Turner, Beckett, and Afton, *Farm Production*; and the price data are from http://www.iisg.nl/hpw/data.php and Clark, "Long March of History."
Method 1: $\varepsilon = -\sigma_q/\sigma_p$ [for explanation, see the text]
Method 2: $\ln(q_t) = a + b.\ln(p_t)$ [for explanation, see the text]

of ln(q) and ln(p) gives a very similar price elasticity of –0.57 for gross wheat yields, indicating that in the thirteenth and fourteenth centuries a doubling of wheat prices meant a harvest shortfall of somewhat less than one-half. These results are consistent with the evidence of the yield data, which confirm that in the worst harvests of all—in the case of wheat, 1316, 1339, 1349, 1350, 1351, 1420, 1438, and 1477—less than three-fifths the trend value of the net yield were returned. Such dismal harvests must have been felt throughout the economy and impacted in different ways upon all social classes. Breaking the data set at the Black Death (1349) reveals no striking difference between the two periods.[57]

Applying the same formula to the Turner, Beckett, and Afton yield data set for the period 1750–1850 produces a much lower elasticity of –0.23. This is well below Davenant's estimate of –0.403 for the seventeenth century and lower too than Barquin's figure of –0.36 for the period 1540–1840. Rising living standards could help explain the reduced demand elasticity for a non-inferior good such as wheat. Fogel's elasticity formula implies that the reduction in crop yield variations discussed above would have operated in the same direction, but that an increase in market integration, by reducing price volatility, would have increased elasticities.[58]

[57] The yield data are from Campbell, "Three Centuries"; and Turner, Beckett, and Afton, *Farm Production*; and the price data is from http://www.iisg.nl/hpw/data.php. In line with Persson, *Grain Markets*, pp. 55–62 (compare Fogel, "Second Thoughts," pp. 252–55), it is assumed here that carry-over stocks of grain were small; England's net foreign trade in grain is also assumed to have been insignificant in this period.

[58] Gras, *English Corn Market*, remains the only study of the grain trade across five centuries.

THE "ULTIMATE CHECK" IN PREINDUSTRIAL ENGLAND

Whereas England was haunted by famine in the Middle Ages, and was probably hit harder by the Great European Famine of 1315–1321 than any subsequent harvest crisis, it had already largely banished that specter by the time it embarked upon its Industrial Revolution and Malthus wrote his *Essay*.[59] What has usually been seen as England's last significant subsistence crisis was confined to the upland north-west of the country in the 1620s and had only a modest national demographic impact.[60] Yet a run of bad harvests again brought England to the brink of famine in 1647–1650 and a case can be made for the recurrence of genuine subsistence problems in both 1727–1729 and 1740–1742.[61] The latter, in fact, was the last occasion in English demographic history when a surplus of deaths over births produced a negative rate of population increase at a time of acute food scarcity. Marriage and birth rates, too, were depressed—a sure sign of economic hardship.[62] Thereafter, although extreme weather events recurred, England, and a growing number of European countries, remained effectively famine-free (with Ireland one of the last to join the club).[63] When, in 1800/01, at the height of the Napoleonic blockade and to the immense concern

[59] Lee, "Short-Term Variation," p. 399, found "a positive but weakening relation between prices and mortality" until 1745; thereafter "the relation becomes negative and very weak." Kelly and Ó Gráda, "Living Standards and Mortality," link the disappearance of the positive check to the introduction of systematic poor relief.

[60] Appleby, *Famine*; and Hoyle, "Famine." See also, Laslett, *World We Have Lost*, pp. 119–25; and Slack, *Poverty and Policy*, pp. 45–46, 48–50.

[61] Hindle, "Dearth"; Campbell, "Four Famines"; and Gooder, "Population Crisis." The 1727–1729 crisis resulted in an excess mortality of about 170,000, or over 3 percent of the population; the 1740–1742 crisis resulted in about 80,000 deaths. Exceptionally for the period, both crises led to grain imports.

[62] Wrigley and Schofield, *Population History*, pp. 531–35. The situation in England was far less serious than in neighboring Ireland, where the 1740–1742 crisis ranks as a major famine, which, on David Dickson's reckoning, may have reduced the population by as much as 15–20 percent: *Arctic Ireland*, p. 69. Environmentally, 1740 is the coldest year on record in Manley's post-1659 series of Central England Temperatures. The years 1740–1742 stand out in dendrochronologies from both the northern and southern hemispheres as years of minimum tree growth, which bears clear testimony to the exceptional climatic conditions then prevailing: Baillie, "Abrupt Environmental Change," p. 56 and "Dendrochronology," pp. 105–07, 112–15, 119; and Post, "Climatic Variability." On the pan-European dimensions of the crisis, see Post, *Food Shortage.*

[63] One consistent and "impartial" measure of annual variations in growing conditions is provided by the variance of detrended ring widths of British Isles oaks: Thirteenth century = 15.3 percent; fourteenth century = 12.3 percent; fifteenth century = 13.7 percent; sixteenth century = 14.2 percent; seventeenth century = 16.5 percent; eighteenth century = 11.6 percent; nineteenth century = 12.6 percent; and twentieth century = 11.5 percent (25-master oak dendrochronology supplied by M. G. L. Baillie). On the crisis of the 1840s, see Ó Gráda, Vanhaute, and Paping, *When the Potato Failed*; and Baillie, "Bad for Trees."

of the government, the English harvest failed and grain prices soared, no major mortality crisis ensued. Instead, the government's emergency measures, local poor relief, and voluntary fertility controls took most of the strain.[64]

The eradication of famine was a signal achievement that contributed materially to the wellbeing of the humblest members of society. Why did it happen? It may have owed something to an exogenous dampening down of the environmental instability that had been such a feature of the fourteenth and fifteenth centuries during the transition from the Medieval Climate Anomaly to the Little Ice Age.[65] Human agency was also material, via the increasing capacities of governments, markets, welfare institutions, and society at large to cope with environmental hazards and their ensuing harvest shortfalls. The more effective functioning of grain markets, as a result of better arbitrage between markets and between good and bad years (a process in which governments themselves were active agents), is well documented, and consistent with the halving in the magnitude of year-to-year variations in the price of wheat which took place between the fourteenth and nineteenth centuries in England, paralleling a similar secular reduction elsewhere in much of western Europe.[66] Year-on-year price changes (measured as absolute values) averaged about 25 percent during the thirteenth and fourteenth centuries, declined to 15–16 percent in the following two centuries, and eventually shrank to 13 percent between the seventeenth and nineteenth centuries.[67] Impressively, much of this was achieved before, first, the canal and, then, the railway and steamship revolutions transformed the economics of transporting and transshipping grain in bulk.

Lower interest rates, by reducing the costs of storage and thereby encouraging larger carryovers, probably aided and abetted the effects of growing market efficiency.[68] Adoption, at the instigation of central government, of a nationwide system of public poor relief following

[64] Wells, *Wretched Faces*; Wrigley and Schofield, *Population History*, p. 534; and Campbell, "Four Famines."

[65] Campbell, "Panzootics."

[66] Fogel, *Escape from Hunger*, pp. 53–54. Persson, *Grain Markets*, pp. 91–113; and Epstein, *Freedom and Growth*, pp. 155–67 (and especially figure 7.3).

[67] The data are taken from Clark, "Price History," appendix table 1. The decline was temporarily halted during the sixteenth century, due to a combination of genuine harvest shocks, particularly in the 1550s and 1590s, and the massive monetary turbulence in the 1540s and 1550s caused by the Great Tudor Debasement (Gould, *Great Debasement*). In 1551 the price of silver in England was six times its 1540 level, although it had hardly changed in Paris or Amsterdam. Since English grain markets were well integrated even in the Middle Ages, long-term trends in the year-to-year variability of local series (for example, Exeter, London) broadly follow that described in the text.

[68] Epstein, *Freedom and Growth*, pp. 60–62; McCloskey and Nash, "Corn at Interest"; Clark, "Cost of Capital"; and Bateman, "Evolution."

the run of bad harvests in the 1590s must also have helped to alleviate distribution problems and protect the food entitlements of the most vulnerable members of English society during crises.[69] Very likely this had the secondary benefit in years of acute scarcity of restraining the pressure on food prices that would otherwise have arisen from inelastic demand. The lack of any pronounced upward demographic dynamic between 1650 and 1750 may also have taken some of the heat out of annual grain price variations, a development reinforced by the rising productivity of domestic agriculture manifest in England's emergence after 1700 as a net grain exporter on a significant scale.[70] The result of these changes was reduced sensitivity of prices to harvests. The magnitude of grain price variations during the "cold" seventeenth century was significantly less than that during the "unstable" fourteenth century, and during the eighteenth and nineteenth centuries it fell still further.[71]

These changes are reflected in the reduction in grain price elasticities documented in this study. Because genuine food-availability declines long remained an inescapable agronomic fact of life, preindustrial European price elasticities of grain were *not* low. In fact, they were 50 percent higher than in England in the era of the Industrial Revolution and twice or three times as high as at the end of the nineteenth century. Progressively and with gathering momentum, reduced storage costs and better access to foreign grain markets had reduced the response of prices to domestic harvest shocks. Given that wheat was not an "inferior" good (in the economic sense), any ensuing increase in incomes will also have reduced demand elasticity. However, the reduction was not all due to improved market integration. This article has also drawn attention to another reason why the risk of famine fell: a reduction in the annual variability of grain harvests. A decline in yield variability of the order indicated above—one-quarter or so between the fourteenth and eighteenth centuries (Table 3)—would have reduced the elasticity of demand by about the same amount. It remains to be established how precisely this was achieved but most likely it was

[69] Hindle, *On the Parish?*; Smith, "Social Security"; Kelly and Ó Gráda, "Poor Law of Old England"; and Healey, "Economic Fluctuations." Slack, *Poverty and Policy*, reckoned that by the end of the seventeenth century, poor law expenditure amounted to *c.*1 percent of national income (sufficient to provide year-round subsistence for 5 percent of the population); spending on poor relief doubled to *c.*2 percent of national income by the end of the eighteenth century.

[70] Ormrod, *English Grain Exports*, pp. 96–97. See also Apostolides et al., "English Agricultural Output." Action taken by Stuart governments to protect grain supplies conceivably also helped dampen annual price variations (Fogel, "Second Thoughts").

[71] See footnote 65 for the lack of any corresponding trend in the variance of British Isles oak growth (between 1270 and 1470 the variance of grain yields and oak growth had broadly tracked each other).

linked to increased specialization according to comparative advantage by individual producers combined with the biological innovation of selecting and sowing sounder seed.[72] Through trial and error and a process of social learning, farmers somehow succeeded in limiting the variability of harvests and, thus, the magnitude of annual price variations.[73]

From the seventeenth century on, major harvest shortfalls in England gradually ceased to present a serious economic and demographic threat and became more a source of public disorder than of excess mortality. The frequent "moral economy" food riots of this later period arose in large part from the growing share of English households lacking any direct involvement in food production and reflected temporary scarcities rather than imminent famine. Agitation, once quelled, was usually accompanied by concessions and conciliation. Meanwhile, a combination of economic development, better governance, and, perhaps, biological innovation were making subsistence more secure. It was paradoxical, therefore, that, as the threat of genuine famine receded, harvest shortfalls grew to assume an importance to civil authorities and political representatives that they had never possessed during the far hungrier Middle Ages.[74]

[72] On the contribution of biological innovations such as the sowing of improved seed varieties to the maintenance and improvement of American wheat yields during the nineteenth century, see Olmstead and Rhode, *Creating Abundance.* For biological evidence of late medieval English field crops, see Letts et al., *Smoke Blackened Thatch.*

[73] Allen, *Enclosure and the Yeoman*; and Overton, *Agricultural Revolution*.

[74] Frank Jr., "'Hungry Gap'"; Randall and Charlesworth, *Moral Economy*; Randall, *Riotous Assemblies*; and Bohstedt, *Politics of Provisions*.

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